

Delayed postglacial colonization of *Betula* in Iceland and the circum North Atlantic

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Abstract

As the Arctic continues to warm, woody shrubs are expected to expand northward. This process, known as “shrubification”, has important implications for regional biodiversity, food web structure, and high-latitude temperature amplification. While the future rate of shrubification remains poorly constrained, past records of plant immigration to newly deglaciated landscapes in the Arctic may serve as useful analogues. We provide one new postglacial Holocene sedimentary ancient DNA (*sedaDNA*) record of vascular plants from Iceland and place a second Iceland postglacial *sedaDNA* record on an improved geochronology; both show Salicaceae present shortly after deglaciation whereas Betulaceae first appears more than 1000 years later. We find a similar pattern of delayed Betulaceae colonization in eight previously published postglacial *sedaDNA* records from across the glaciated circum North Atlantic. In nearly all cases, we find that Salicaceae colonizes earlier than Betulaceae and that Betulaceae colonization is increasingly delayed for locations farther from glacial-age woody plant refugia. These trends in Salicaceae and Betulaceae colonization are consistent with the plant families’ environmental tolerances, species diversity, reproductive strategies, seed sizes, and soil preferences. As these reconstructions capture the efficiency of postglacial vascular plant migration during a past period of high-latitude warming, a similarly slow response of some woody shrubs to current warming in glaciated regions, and possibly non-glaciated tundra, may delay Arctic shrubification and future changes in the structure of tundra ecosystems and temperature amplification.

eLife assessment

This **valuable** work on the paleovegetation history of Iceland has implications for the field of paleoecology, and the deglaciation history of Iceland and additional localities in Northern America and Europe via woody shrub colonization. The study uses a sedimentary ancient DNA metabarcoding approach to study this historic process. The strength of evidence is **solid**, with the methods (analysis of sedimentary DNA) and data analyses broadly supporting the claims.

Introduction

As the planet warms, the Arctic is warming at 2 to 4 times the global rate (Rantanen et al., 2022). This warming is impacting Arctic ecosystems through a process known as ‘shrubification’, or the northward expansion and increased height and density of deciduous woody vegetation (Tape et al., 2006; Myers-Smith et al., 2011; Elmendorf et al., 2012; Sweet et al., 2015). Arctic shrubification alters the structure of tundra ecosystems, its regional biodiversity, food web structure, and nutrient availability (Elmendorf et al., 2012; Fauchald et al., 2017; Collins et al., 2018). Arctic shrubification also has implications for the climate system by reducing surface albedo (Sturm et al., 2005) and increasing atmospheric water vapor (Pearson et al., 2013) – both positive feedbacks that amplify high-latitude warming (Thompson et al., 2022). However, large uncertainties remain regarding the rate of future shrubification, such that the magnitude of these positive feedbacks is poorly constrained in predictive earth system models. One way to reduce future climate uncertainties is through the analysis of well-constrained paleoenvironmental records dating to warmer-than-present periods in Earth’s recent history (e.g., Tierney et al., 2020).

Traditionally, macrofossils and pollen have formed the backbone of Quaternary paleovegetation reconstructions (Birks, 2018). However, macrofossils are not consistently preserved in sedimentary records, and pollen production is restricted to seed-bearing taxa, which can be obscured by long-distant transport from sources 1000s of km away (Hyvärinen, 1970; Birks, 2003; Crump et al., 2019). For example, in Iceland, exotic tree pollen from taxa never reported as macrofossils since the Miocene and Pliocene (i.e., *Pinus*, *Quercus*, *Corylus*, *Alnus*, and *Ulmus*, Denk et al., 2011a, b) are found in Early Holocene sedimentary records following retreat of the Icelandic Ice Sheet (Caseldine et al., 2006; Eddudóttir et al., 2015). Hence, the first appearance of pollen need not reflect local colonization of a given taxon. Recent analytical advances in sedimentary ancient DNA (*sedaDNA*) allow for more reliable vegetation reconstructions (e.g., Capo et al., 2021), due in part to their application in continuous lake sediment records that capture colonization times following local deglaciation. Taxa identified by DNA in lake surface sediment closely match those in catchment vegetation cover, confirming the local origin of DNA in lake sediment and improved reliability for taxa presence compared to pollen (Sjögren et al., 2017; Alsos et al., 2018). Past records of vascular plant *sedaDNA* can therefore provide new insight into postglacial colonization patterns (Crump et al., 2019; Alsos et al., 2022) as well as the response of vegetation to changes in temperature during prior warm periods (Clarke et al., 2020; Crump et al., 2021; Huang et al., 2021).

In this study we present one new Icelandic record (Stóra Viðarvatn) and update the age model of a second previously published record from Iceland (Torfdalsvatn, Alsos et al., 2021). For these two lakes, we also compare existing pollen and *sedaDNA* datasets to highlight the ability of *sedaDNA* in pinpointing the timing of each taxon’s first appearance. We then compare *sedaDNA* records from eight other published North Atlantic lakes that track local vegetation assemblages since deglaciation, with particular attention to postglacial colonization patterns of two woody plant families, Salicaceae and Betulaceae. Collectively, the North Atlantic vascular plant DNA data sets demonstrate that Salicaceae colonized deglaciated lands quickly, whereas Betulaceae was delayed by up to several thousand years. As these colonization patterns took place during a period of high latitude warming, our datasets suggest that ongoing warming may also lead to different northward expansion rates of woody shrubs in glaciated regions, and possibly tundra regions as well. As a result, the rate of Arctic shrubification may be delayed, with important consequences for high-latitude ecosystems and temperature amplification.

Stóra Viðarvatn, Iceland

Stóra Viðarvatn (#4, [Table 1](#), [Figure 1](#)) is a large lake (surface area: 2.4 km², depth: 48 m) in northeast Iceland, with no pre-existing records of Holocene vegetation change. In winter 2020, we recovered sediment core 20SVID-02 from 17.4 m water depth near the center of the lake. The core contains 893 cm of sediment, collected in 1.5 m increments. The basal 20 cm are laminated, clay-rich sediment, transitioning to organic gyttja by 873 cm depth, which characterizes the remainder of the record, interbedded with tephra (volcanic ash) deposits ([Figure S1](#)). The chronology of the Early to Middle Holocene portion of the record is based on 5 tephra layers (see Materials and Methods), of which the lowest, at 886.5 cm depth, is identified as the Askja S tephra (10830 ± 57 BP, [Bronk Ramsey et al., 2015](#)), and the highest, at 334 cm depth, as Hekla 4 (4200 ± 42 BP, [Dugmore et al., 1995](#)). Because the sedimentation rate between the six tephra layers is essentially linear, we extrapolate that rate from the Askja S tephra layer 7 cm to the base of the sediment core at 893.5 cm depth, which provides a minimum deglaciation age of ~10950 BP ([Figure S2A](#)).

Samples for *sedaDNA* analysis were taken from 75 equi-spaced depths throughout core 20SVID-02. Of these, 73 yielded amplifiable vascular plant DNA using the *trnL* P6 loop primer set. The two samples that failed (810 and 856.5 cm depth) were sampled within thick tephra layers with low organic content. Following data filtering (see Materials and Methods), the *trnL* dataset yields 15,738,794 total assigned reads, with an average of 209,851 reads per sample. qPCR cycle threshold (C_T) values, which reflect PCR efficiency and the quantity of suitable target sequences for amplification, average 32.1 across samples. While there is inter-sample variation in DNA reads and C_T values, linear regressions show nearly stable temporal trends ([Figure S2](#)), which indicates that the availability of DNA to PCR amplification is consistent throughout the record. Metabarcoding technical quality (MTQ) and analytical quality (MAQ) scores are below suggested thresholds for low quality (0.75 and 0.1, respectively, [Rijal et al., 2021](#)) only in some samples during the Early Holocene ([Figure S2](#)). However, given that the scores correlate with species richness due to the latter values all falling below 30 ([Figure S2](#)), the low MTQ and MAQ scores are likely an artifact of the requirement that the 10 best represented barcode sequences are required for calculation ([Rijal et al., 2021](#)), and not an indication of low-quality DNA. As PCR amplification results in sequence read abundances that may not reflect original relative abundances in a sample ([Nichols et al., 2018](#)), we focus our discussion on taxa presence/absence, with a greater number of replicates representing a higher likelihood of taxa presence. We identified 53 taxa throughout Stóra Viðarvatn's lake sediment record. Of the key woody shrub taxa, Salicaceae is present at the base of the core (892.5 cm depth, 10850 BP) and is present nearly continuously through the core. Betulaceae first appears ~2300 years later (701.5 cm depth, 8560 BP), and has a sporadic presence until the Middle Holocene at ~5500 BP ([Figure 2A](#)).

Torfdalsvatn, Iceland

Torfdalsvatn (#3, [Table 1](#), [Figure 1](#)) is a small lake (surface area: 0.41 km², depth: 5.8 m) in north Iceland from which sediment cores have been analyzed since the early 1990s to create paleoecological records from pollen and macrofossils ([Björck et al., 1992](#); [Rundgren, 1995](#), [1998](#)), and *sedaDNA* ([Alsos et al., 2021](#)). The paleovegetation records derived from pollen/macrofossils and *sedaDNA* were constructed from separate sediment cores that have independent age models derived from radiocarbon and tephra layers of known age. To temporally compare the two sediment records, radiocarbon ages need to be calibrated using the same calibration curve, correlative tephra layer ages must agree, and age model construction requires a common method (i.e., linear interpolation vs. Bayesian). For the pollen/macrofossil records ([Rundgren 1995](#), [1998](#)), this required calibrating the radiocarbon ages using IntCal20 ([Reimer et al., 2020](#)) and application of a Bayesian age modeling approach that informs sediment rate changes with prior information ([Blaauw and Christen, 2011](#)) (see [Geirsdóttir et al., 2020](#)). For

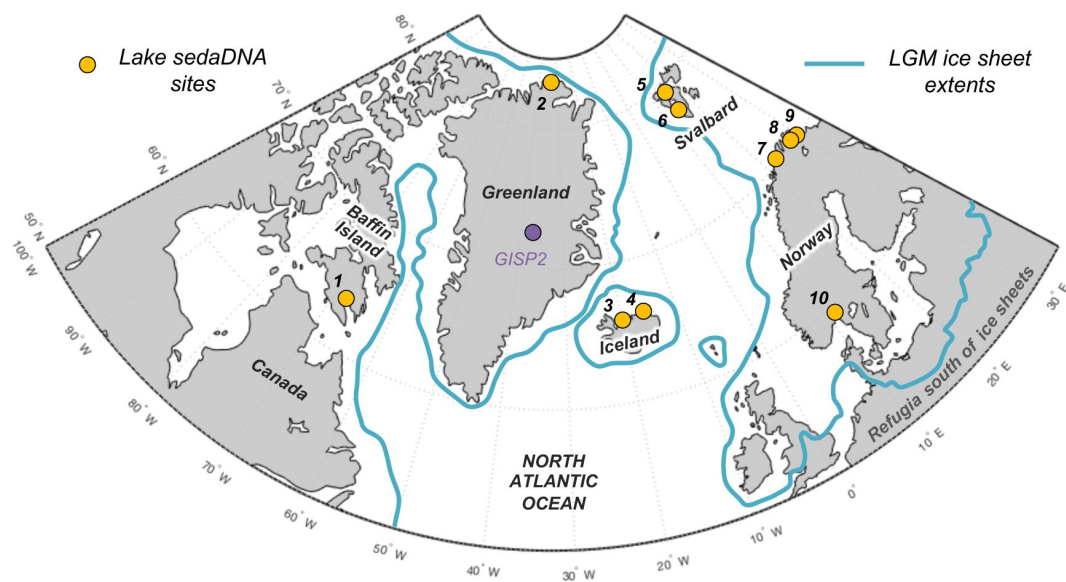


Figure 1

Overview map of the North Atlantic. Lake *sedaDNA* sites with vascular plant histories since local deglaciation shown in yellow (see [Table 1](#) for site information). The extent of the Last Glacial Maximum (LGM) ice sheets is delineated in blue (Batchelor et al., 2019).

Site #	Lake name	Region	Latitude (°)	Longitude (°)	Reference
1	Lake Qaupat	Baffin Island	63.68	-68.20	Crump et al. (2019)
2	Bliss Lake	Greenland	83.52	-28.35	Epp et al. (2015)
3	Torfdalsvatn	Iceland	66.06	-20.38	Alsos et al. (2021)
4	Stóra Viðarvatn	Iceland	66.24	-15.84	This study
5	Jodavannet	Svalbard	77.34	16.02	Volstad et al. (2020)
6	Lake Skartjørna	Svalbard	77.96	13.82	Alsos et al. (2016)
7	Langfjordvannet	Norway	70.15	20.54	Alsos et al. (2022)
8	Eaštorjávri South	Norway	70.43	27.33	Alsos et al. (2022)
9	Nordvivatnet	Norway	70.13	29.01	Alsos et al. (2022)
10	Lake Ljøgottjern	Norway	60.15	11.14	ter Schure et al. (2021)

Table 1

Lake site information.

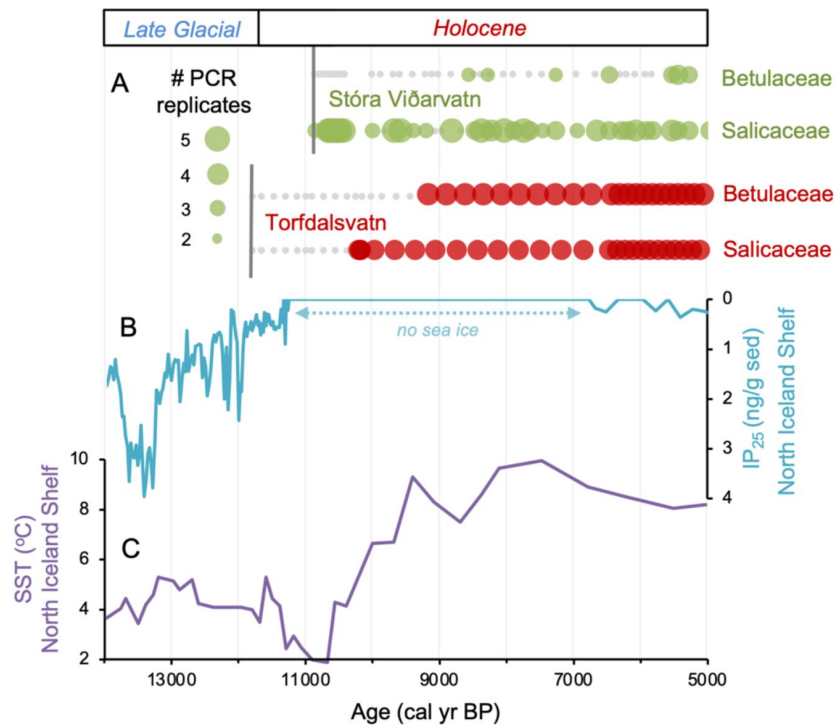


Figure 2

Comparison of paleoclimate and lake *sedaDNA* records from Iceland. A) Presence/absence *sedaDNA* record from Stóra Viðarvatn (green) and Torfdalsvatn (red) for Betulaceae and Salicaceae, where light gray markers denote that no taxa was detected. For Stóra Viðarvatn *sedaDNA*, the bubble size is proportional to the number of PCR replicates, and vertical dark gray lines denote the timing of each lake's deglaciation. B) Biomarker-proxy record of sea ice from the North Iceland Shelf (inverted), where greater concentrations of IP₂₅ correspond to more sea ice (Xiao et al., 2017 [DOI](#)). C) Diatom-based SST record from the North Iceland Shelf (Sha et al., 2022 [DOI](#)).

the pollen/macrofossil records (Rundgren 1995 [↗](#), 1998 [↗](#)), this required calibrating the radiocarbon ages using IntCal20 (Reimer et al., 2020 [↗](#)) and application of a Bayesian age modeling approach that informs sediment rate changes with prior information (Blaauw and Christen, 2011 [↗](#)) (see Geirsdóttir et al., 2020 [↗](#)). For the *sedaDNA* record (Alsos et al., 2021 [↗](#)), we revised the tephra ages previously used to establish a more reliable Bayesian age model (see Methods and Materials). In doing so, the *sedaDNA* and pollen records feature similar linear sedimentation rates and are now synchronized by the G10ka Series tephra layers (Figure S2C). The revised age model for the Early Holocene portion of the pollen/macrofossil record (Figure S2B, Geirsdóttir et al., 2020 [↗](#)) sets the timing of deglaciation, inferred from the onset of organic sedimentation at 801 cm depth, to 11800 BP.

Differences on the order of centuries between the pollen/macrofossil and *sedaDNA* records may be artifacts of chronological uncertainty. This is because although the revised age models for the pollen/macrofossil and *sedaDNA* records feature common tephra layers, variable levels of proxy data resolution between the two records increases uncertainty when comparing the timing of inferred changes. Pollen of both Salicaceae (*Salix spp.*) and Betulaceae (*Betula spp.*) first appear by ~11800 BP (Figure 3 [↗](#)). The oldest *B. nana* macrofossil dates to ~9300 BP, similar to macrofossils of *S. herbacea* (~9400 BP) and *S. phylicifolia* (~9300 BP) (Figure 3 [↗](#)). The *trnL* primed *sedaDNA* data indicate that Salicaceae did not colonize Torfdalsvatn's catchment until ~10300 BP, whereas Betulaceae arrived later at ~9500 BP (Figure 3 [↗](#)).

Resolving Icelandic records of woody taxa colonization

Based on the original age model, Torfdalsvatn pollen records were interpreted to capture ecological changes associated with the abrupt climate oscillations between the Bølling-Allerød (14700 to 12900 BP) and Younger Dryas (12900 to 11700 BP) periods (Björck et al., 1992 [↗](#); Rundgren, 1995 [↗](#)). However, recent calibration of the radiocarbon ages in the Torfdalsvatn cores from Björck et al. (1992) [↗](#) and Rundgren (1995) suggest that the organic-rich portion of the record began less than 11800 years ago (Figure S2B, Geirsdóttir et al., 2020 [↗](#)). While there is sediment at deeper levels, it is described as silty clay (Björck et al., 1992 [↗](#)) with sand and gravel (Rundgren, 1995 [↗](#)). These characteristics suggest that the sediment likely originates from a rapidly deglaciating environment, perhaps with lingering ice sheet meltwater in the catchment and a rapid sediment accumulation rate that is difficult to constrain without secure age control. Therefore, we suggest that the maximum age for the final retreat of the Icelandic Ice Sheet from Torfdalsvatn's catchment, when organic-rich sedimentation began, is ~11800 BP, several thousand years younger than previously proposed (Björck et al., 1992 [↗](#); Rundgren, 1995 [↗](#)). We use the updated age models to reconstruct the history of plant colonization in Iceland.

Differences in taxa presence/absence between the *sedaDNA* and pollen records from Torfdalsvatn likely reflect different mechanisms of transport for DNA and pollen. *SedaDNA* for both Salicaceae and Betulaceae from Torfdalsvatn shows a colonization delay relative to deglaciation, whereas pollen from both taxa is present, albeit in low abundance, in the oldest sediment samples (Figure 3 [↗](#)). This is most likely explained by the local catchment origin of *sedaDNA* (Sjögren et al., 2017 [↗](#)) versus pollen, which can be transported by wind from 1000s of km away (Caseldine et al., 2006 [↗](#); Eddudóttir et al., 2015 [↗](#)). Therefore, the occurrence of low pollen counts prior to the corresponding identification with *sedaDNA* (Alsos et al., 2021 [↗](#)) likely reflects long-distance transport from sources outside of Iceland rather than local establishment in the catchment – an inference supported by a lack of *Salix* and *Betula* macrofossils during this interval (Figure 3 [↗](#), Rundgren, 1998 [↗](#)). While poor DNA preservation could explain the delay in *sedaDNA*, other vascular plants are identified before Salicaceae and Betulaceae DNA appear, suggesting adequate DNA preservation for PCR amplification (Alsos et al., 2021 [↗](#)). Moreover, the corresponding sediment is rich in clay (Rundgren, 1995 [↗](#)), which preferentially binds with and stabilizes DNA

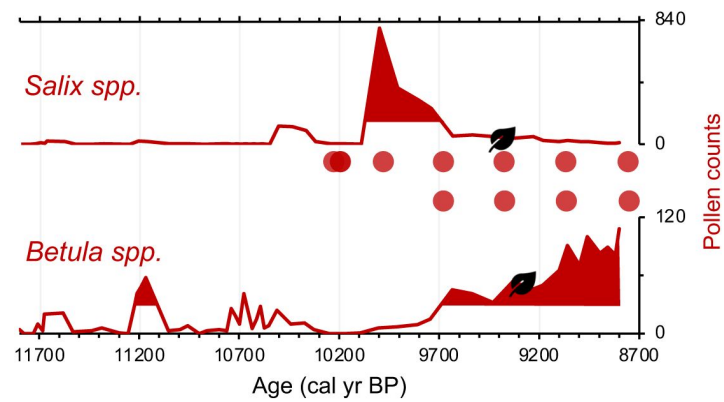


Figure 3

Simplified paleovegetation records from Torfdalsvatn for two taxa: *Salix spp.* and *Betula spp.* Shown are pollen counts (bold red lines), where shaded regions indicate values above the mean (Rundgren, 1995 [link](#)), first occurrence of taxa macrofossils (black leaves, Rundgren, 1998 [link](#)), and DNA presence (red bubbles, Alsos et al., 2021 [link](#)).

(Kanbar et al., 2020 [↗](#)), making poor preservation unlikely. Therefore, we conclude that the colonization of Salicaceae and Betulaceae were delayed by 1500 and 2300 years, respectively, after local deglaciation of Torfdalsvatn's catchment.

Relative to Torfdalsvatn, the catchment of Stóra Viðarvatn deglaciated ~1000 years later at ~10850 BP (**Figure 2A** [↗](#)). Based on our new *sedaDNA* record, Salicaceae colonized the catchment rapidly, as it was identified in the lowermost sample at ~10850 BP, whereas Betulaceae does not appear in the record until 8560 BP, more than 2000 years later. Considering CT values are relatively stable throughout the entire Holocene record (Figure S3), the delayed appearance of Betulaceae does not likely result from preservation bias. While Stóra Viðarvatn does not have any independent paleovegetation records, a pollen record from a nearby peat section Ytra-Áland in Thistilfjörður (Karlsdóttir et al., 2014 [↗](#)), ~12 km to the east, provides a useful cross-examination of woody shrub colonization patterns. In the Ytra-Áland record, *Salix* pollen is found in high counts well above the mean in the lowermost sample dated to 10400 BP, suggesting bona fide early colonization (**Figure 4** [↗](#)). For *Betula*, there is limited to no pollen present until ~8000 BP, after which there is sporadic presence as we observe with *sedaDNA* (**Figure 4** [↗](#)). The *Betula* pollen-limited period at Ytra-Áland is followed by the first peak of *Betula* presence in the total land pollen during ~7700 to 6800 BP. According to size differentiation between the two *Betula* species in Iceland (Karlsdóttir et al., 2007), the pollen recovered in this peak period is mostly from the dwarf birch *B. nana*, but with a fair amount of pollen from the shrub-like *B. pubescens*, suggesting arctic-alpine heath with stands of downy birch in the region (Karlsdóttir et al., 2014 [↗](#)). Despite the coarser resolution of Ytra-Áland's age model, which only has two tephra layer dates bounding the timing of first appearance of these taxa (10400 BP and 7050 BP, Karlsdóttir et al., 2014 [↗](#)), there is reasonable agreement between lake and peat records for the timing of arrival and subsequent Early Holocene establishment of Salicaceae and Betulaceae in northeast Iceland.

Based on the *sedaDNA* records from Torfdalsvatn and Stóra Viðarvatn, we find that as the Icelandic Ice Sheet receded and exposed new land Salicaceae colonized the landscape. This occurred 0 to 1700 years after inferred deglaciation, whereas Betulaceae colonization was further delayed by a few thousand years (~2300 years in both cases). As Torfdalsvatn is the oldest dated terrestrial site of deglaciation in Iceland, this implies that ~1700 years was required for Salicaceae to establish a viable population on the island. By the time Stóra Viðarvatn deglaciated, it is likely that other portions of north Iceland had also deglaciated allowing Salicaceae to more efficiently colonize new regions of Iceland. The similar timing of Betulaceae colonization relative to deglaciation in the two catchments (~2300 years) but different times of deglaciation (11800 and 10850 BP) suggests that the evolution of catchment conditions for Torfdalsvatn and Stóra Viðarvatn may be additional factors controlling the local colonization of this taxon on Iceland. The earlier presence of low concentrations of Salicaceae and Betulaceae pollen in Torfdalsvatn are most likely derived from long distance wind dispersal from outside Iceland.

Discussion

Postglacial *sedaDNA* records from the circum North Atlantic

Eight additional records of vascular plant *sedaDNA* from lake sediments located around the North Atlantic that were glaciated during the Last Glacial Maximum are available to compare with the Icelandic data (see Material and Methods for details). These include records from Baffin Island ($n=1$, Crump et al., 2019 [↗](#)), northern Greenland ($n=1$, Epp et al., 2015 [↗](#)), Svalbard ($n=2$, Alsos et al., 2016 [↗](#); Volstad et al., 2020 [↗](#)), northern Norway ($n=3$, Rijal et al., 2021 [↗](#); Alsos et al., 2022 [↗](#)), and southern Norway ($n=1$, ter Schure et al., 2021) (**Figure 1** [↗](#)). All eight records have age control derived from calibrated radiocarbon ages. Except for Baffin Island, we infer minimum ages of deglaciation based on the age of the oldest sediment.

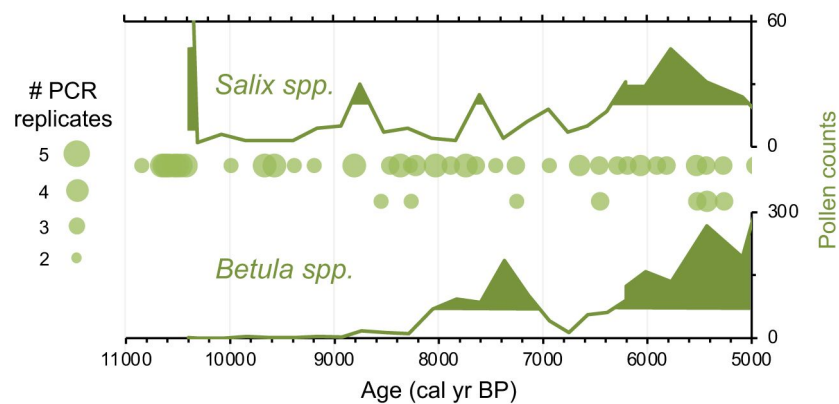


Figure 4

Simplified paleovegetation records from northeast Iceland for two taxa: *Salix spp.*, and *Betula spp.* Shown are Ytra-Áland pollen counts (bold green lines), where shaded regions indicate values above the mean (Karlsdóttir et al., 2014) and DNA presence from Stóra Viðarvatn (green bubbles, this study), where the bubble size is proportional to the number of PCR replicates.

Baffin Island: Lake Qaupat (#1, [Figure 1](#)) is a small lake (surface area: 0.08 km², depth: 9.2 m) situated at 35 m asl on southern Baffin Island. Cosmogenic ¹⁰Be exposure dating of Lake Qaupat's impounding moraine constrains the timing of deglaciation to 9100 ± 700 BP (Crump et al., 2019). Lake Qaupat and most of its catchment were below sea level until 7700 ± 300 BP, when postglacial isostatic recovery raised the basin above the ocean and its catchment was available for vascular plant colonization. Salicaceae is present in the record at shortly after isolation from the ocean at 7400 BP, but no vascular plants are documented by *seda*DNA in the older marine sediment. The first appearance of Betulaceae *seda*DNA occurs at 5900 BP, more than 1000 years after Salicaceae and 2000 years after the lake and its catchment were above sea level (Crump et al., 2019).

Greenland: Bliss Lake (#2, [Figure 1](#)) is a small lake (depth: 9.8 m) situated at 17 m asl on the northern coastline of Greenland (Peary Land). Bliss Lake initially deglaciated 11000 BP (Olsen et al., 2012) and the first appearance of Salicaceae *seda*DNA is found in low read counts in sediment dated to 10800 BP but only in high read counts beginning at 7400 BP (Epp et al., 2015). As the lake was inundated by marine water between 10480 and 7220 BP (Olsen et al., 2012), 7400 BP is interpreted to be the most reliable date for Salicaceae colonization in the catchment as sea level regressed and isolated the lake from the ocean (Epp et al., 2015).

Svalbard: Jodavannet (#5, [Figure 1](#)) is a small lake (depth: 6.4 m) situated at 140 m asl on the east coast of Wijdefjorden on northern Spitsbergen. Regional cosmogenic exposure dating suggests the area began to deglacierate between 14600 and 13800 BP (Hormes et al., 2013), although the base of Jodavannet's sediment record suggests a minimum timing of lake deglaciation by 11900 BP (Volstad et al., 2020). Salicaceae *seda*DNA is first found at 9800 BP in Jodavannet, whereas Betulaceae was not detected with *seda*DNA metabarcoding (Volstad et al., 2020). Lake Skartjørna (#6, [Figure 1](#)) is a small lake (surface area: 0.10 km², depth: 7.5 m) situated at 65 m asl on the west coast of Spitsbergen. As sea level fell due to glacial isostatic adjustment, Lake Skartjørna likely became isolated from the sea at ~13000 BP (Landvik et al., 1987), although the base of the recovered sediment record, which provides a minimum deglaciation age, is dated to 8600 BP (Alsos et al., 2016). Salicaceae *seda*DNA is identified in Lake Skartjørna's basal sample at 8600 BP and Betulaceae is first identified at 7000 BP (Alsos et al., 2016).

Norway: Langfjordvannet (#7, [Figure 1](#)) is a small lake (surface area: 0.55 km², depth: 34.8 m) situated at 66 m asl on the coast of northern Norway (Rijal et al., 2021). Langfjordvannet deglaciated by 16150 BP, Salicaceae is first identified in *seda*DNA by 15500 BP, and Betulaceae by 10000 BP (Alsos et al., 2022). Eaštorjávri South (#8, [Figure 1](#)) is a small lake (surface area: 0.06 km², depth: 5.4 m) situated at 260 m asl near the coast of northern Norway (Rijal et al., 2021). Eaštorjávri South deglaciated by 11060 BP, with Salicaceae *seda*DNA identified in the basal sample at 11060 BP and consistently present above that level. Betulaceae is identified, albeit with low read counts, in the basal sample as well, but not identified in the overlying sample. Since Betulaceae *seda*DNA is not present with substantial read counts until 9950 BP (Alsos et al., 2022), we conservatively set this as Betulaceae's first appearance in Eaštorjávri South. Nordvivatnet (#9, [Figure 1](#)) is a small lake (surface area: 0.05 km², depth: 13.3 m) situated at 82 m asl on the coastline of northern Norway (Rijal et al., 2021). Nordvivatnet deglaciated by 12880 BP, with Salicaceae *seda*DNA identified in the basal sample. Betulaceae *seda*DNA is identified in the basal sample as well but below our cut-off for taxa presence (see Materials and Methods). The first sample with high read counts indicative of Betulaceae presence is at 12080 BP (Alsos et al., 2022), which we conservatively set as the timing of the taxa's first appearance in Nordvivatnet. Finally, Lake Ljøggottjern (#10, [Figure 1](#)) is a small lake (surface area: 0.02 km², depth: 18 m) situated at 185 m asl in southeastern Norway. Lake Ljøggottjern deglaciated by 9300 BP and both Salicaceae and Betulaceae *seda*DNA were first detected by 7900 BP (ter Schure et al., 2021).

Betulaceae colonization is delayed relative to Salicaceae in the circum North Atlantic

The timing of postglacial Salicaceae and Betulaceae colonization varies across the North Atlantic regions, with first-appearance dates ranging from 15500 BP in Norway to 5900 BP on Baffin Island (**Figure 5**). Salicaceae appears immediately after inferred deglaciation at 4 out of 10 sites, while colonization dates for the other 6 locations range from 200 to 2100 years after deglaciation and exhibit no clear spatio-temporal pattern (**Figure 6**). For Betulaceae, colonization times range from 800 to 6150 years after deglaciation, with sites closest to source refugia south of the ice sheets (**Figure 1**) having generally shorter colonization delays (**Figure 6**), consistent with seedling viability having a higher probability of success with shorter distance from source (Nathan, 2006). The one exception is Langfjordvannet, in northern Norway (site #7, **Figure 1**, Alsos et al., 2022), which deglaciated at 16150 BP (**Figure 5**). Considering that Betulaceae only arrives during the Holocene Epoch (last 11700 years) across all North Atlantic sites, the anomalously long delay of Betulaceae to Langfjordvannet suggests that this taxon may have required the Holocene's relatively stable climate, as reflected by Greenland oxygen isotope records (**Figure 5**, Seierstad et al., 2014), to successfully establish itself. Alternatively, the colonization delay of Betulaceae to Langfjordvannet may reflect the slow development of stable soils, which are required for Betulaceae establishment, due to the lake's steep catchment (Otterå, 2012). In contrast to Betulaceae, Salicaceae appears to have been an efficient colonizer to Langfjordvannet during the colder and less stable climate of the Late Glacial period and earliest Holocene (**Figure 5**, Seierstad et al., 2014). Finally, except for Lake Ljøgottjern in southern Norway (site #10, **Figure 1**) where Betulaceae and Salicaceae apparently colonize at about the same time (ter Schure et al., 2021), all other sites show that Betulaceae colonization is delayed by 800 to 5500 years relative to Salicaceae colonization dates (**Figure 5**).

Several possibilities may contribute to the delay in Betulaceae colonization compared to Salicaceae colonization. First, the northernmost Salicaceae species reach bioclimate subzone A (1-3 °C July temperature) whereas the northernmost Betulaceae species rarely go beyond bioclimate subzone D (8-9 °C July temperature) (Walker et al., 2005). The different environmental tolerances may explain why Salicaceae generally arrives earlier when regional temperatures are still climbing compared to the timing of Betulaceae colonization when warmth is more broadly established (**Figure 5**). In support, Alsos et al. (2022) showed that at least for Fennoscandian *sedaDNA* vascular plant records, temperature was one of the dominant controls on postglacial dispersal patterns. Second, species diversity is greater for the Salicaceae family compared to Betulaceae family in the North Atlantic, and conventional DNA metabarcoding techniques cannot identify *sedaDNA* to the species level. Iceland, for example, has four native Salicaceae taxa, i.e., three dwarf, cold-tolerant arctic species (*S. herbacea*, *S. lanata*, and *S. arctica*) and one shrub-like species (*S. phylicifolia*), and only two Betulaceae (*B. nana* and *B. pubescens*) (Kristinsson, 2008). The suite of Salicaceae taxa can collectively tolerate a wider range of environments than Betulaceae, including disturbed, wet, and nutrient limited substrate characteristic of deglacial environments (e.g., *S. herbacea*, Beerling, 1998). Third, the modes of reproduction for Salicaceae and Betulaceae are different. Salicaceae is clonal, meaning that plants mostly reproduce asexually by vegetative spreading, which is advantageous when colonizing new landscapes (e.g., Beerling, 1998). Betulaceae propagates instead by seed and therefore requires a larger number of founder individuals for successful establishment (Atkinson, 1992; de Groot et al., 1997). Moreover, Betulaceae seeds are generally heavier than Salicaceae seeds (dry mass, *B. nana* = 0.12 to 0.30 mg, de Groot et al., 1997; *S. herbacea* = 0.07 to 0.14 mg, Beerling, 1998). This mass difference would provide a higher probability of long-distance wind dispersal for Salicaceae compared to Betulaceae and establishment of the pioneer plants.

Finally, modern observations broadly replicate the records of delayed woody taxa colonization in the North Atlantic. Studies tracking the migration of Salicaceae and Betulaceae in the forefields of retreating glaciers and to new islands also show a rapid and faster colonization of Salicaceae

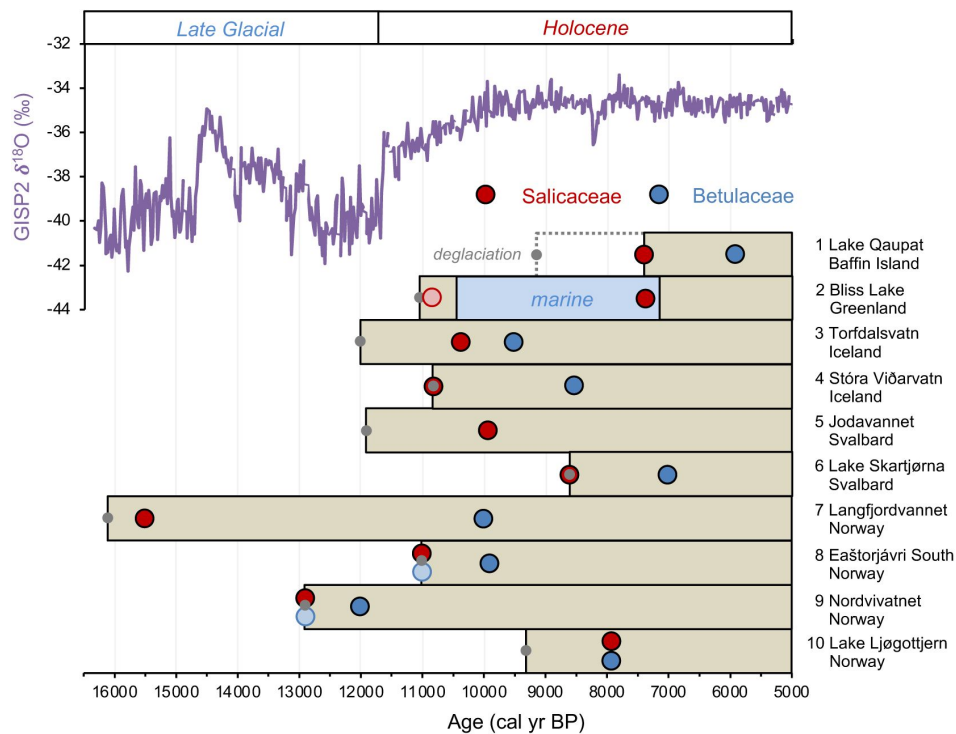


Figure 5

Timing of postglacial Salicaceae and Betulaceae colonization in the circum North Atlantic at 10 locations (see [Figure 1](#) and [Table 1](#)). A) GISP2 $\delta^{18}\text{O}$ reflective of regional North Atlantic temperature variability ([Seierstad et al., 2014](#)), and B) lake sedaDNA records of vascular plants, where the brown boxes reflect the extent of sedimentary record and the red/blue circles denote the first appearance of Salicaceae/Betulaceae ([Epp et al., 2015](#); [Crump et al., 2019](#); [Alsos et al., 2016, 2021](#); [Volstad et al., 2020](#); [Rijal et al., 2021](#); [ter Schure et al., 2021](#); this study). Light red and blue circles denote the presence of a low number of taxa reads, which we conservatively do not interpret to reflect genuine presence (e.g., [Epp et al., 2015](#)). Grey circles denote the minimum timing of local deglaciation inferred from basal core ages, and in the case of Baffin Island, cosmogenic radionuclide dating of the lake's impounding moraine ([Crump et al., 2019](#)).

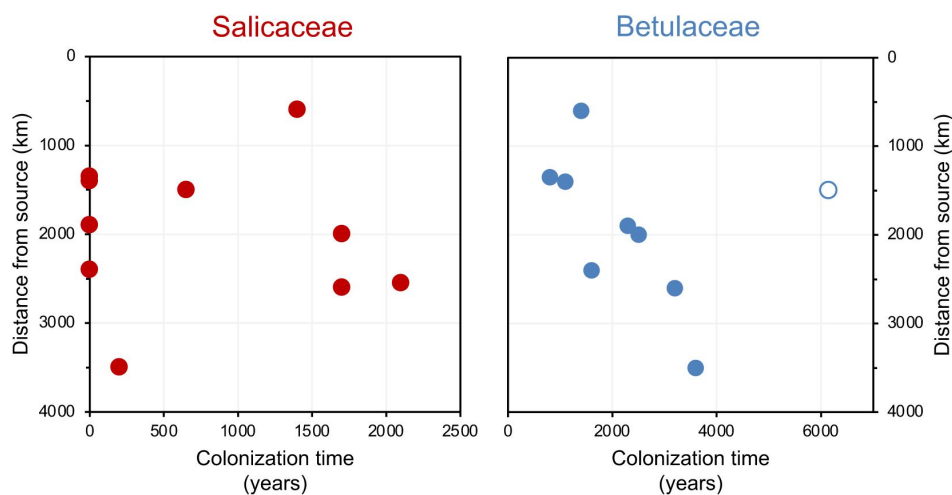


Figure 6

Colonization time after deglaciation (years) versus distance the lake is from the closest possible source south of the LGM ice sheet margin (km). For all circum North Atlantic sites, Salicaceae is shown in red and Betulaceae in blue. Data point for Langfjordvannet is open for Betulaceae.

compared to Betulaceae (Whittaker, 1993 [↗](#); Magnússon et al., 2009 [↗](#); Burga et al., 2010 [↗](#); Synan et al., 2021 [↗](#)). As an example, following the formation of the volcanic island Surtsey in 1964 CE, Salicaceae (*S. herbacea*, *S. lanata* and *S. phylicifolia*) colonized between 1995 and 1999 CE and Betulaceae has yet to arrive (Magnússon et al., 2009 [↗](#)). Complimentary studies tracking soil and nutrient development at proglacial sites in the North Atlantic (Vilmundardóttir et al., 2014 [↗](#)) suggest that Salicaceae is environmentally unconstrained (Whittaker, 1993 [↗](#); Glausen and Tanner, 2019 [↗](#)), whereas Betulaceae requires more developed soil for its ultimate viability (Synan et al., 2021 [↗](#)). Collectively, these studies support the environmental constraints known for each woody taxon and reinforce the hypothesis that Salicaceae is a more efficient pioneer than Betulaceae, particularly in newly deglaciated landscapes.

Environmental dispersal mechanisms

Coupling the timing of postglacial woody shrub colonization in the North Atlantic with genetic approaches provides a valuable opportunity to identify probable environmental dispersal mechanisms (e.g., wind, sea ice, driftwood, and birds) for long-distance transport. Previously, Alsos et al. (2015) [↗](#) compared amplified fragment length polymorphism (AFLP) data from living plants from Iceland, Greenland, Svalbard, the Faroe Islands, and Jan Mayen to AFLP data from the same species from potential source regions. Genetic distances suggest that *Betula* (*sp. nana* and *pubescens*) and *Salix herbacea* populations in most locations descend from ancestors in a western European Last Glacial Maximum (LGM) refugia (Alsos et al., 2009 [↗](#); Eidesen et al., 2015 [↗](#)), and that each taxa underwent a broad-fronted and rapid migration to newly deglaciated regions (Alsos et al., 2015 [↗](#)). *B. nana* on Svalbard likely descends from a northwest Russian source (Alsos et al., 2007, 2015 [↗](#)), and populations on Baffin Island and western Greenland originated from North America (Alsos et al., 2009 [↗](#); Eidesen et al., 2015 [↗](#)). Molecular analysis of Icelandic *Betula* based on chloroplast haplotypes indicates that *Betula* may have post-glacially immigrated to Iceland from northern Scandinavia (essentially *B. nana*) and northwestern Europe (*B. pubescens*) (Thórsson et al., 2010). However, the relative roles of wind, sea ice, driftwood, and birds as transport vectors remain an open question.

Sea ice has been suggested to be a dispersal mechanism for postglacial vascular plants in the Arctic (Alsos et al., 2016 [↗](#)). While seasonal sea ice persisted throughout the Holocene around Svalbard (Pieńkowski et al., 2021 [↗](#)) and northern Greenland (Syring et al., 2020 [↗](#)), general ocean circulation patterns require source populations of vascular plants in Arctic Siberia, the dominant source region for sea ice export to the North Atlantic, for sea ice to be a viable vector. Considering that *B. nana* on Svalbard likely descends from a Russian refugia (Alsos et al., 2007, 2015 [↗](#)), sea ice is a reasonable transport mechanism for this location. However, genetic data from Salicaceae (*S. herbacea*) on Baffin Island, Greenland, and Svalbard suggest that the modern populations from these locations descend from source populations in either mid-latitude North America (south of the Laurentide Ice Sheet) or western Europe (Alsos et al., 2007, 2009 [↗](#)), and neither region exported sea ice in the Early Holocene. This leaves wind and birds as likely mechanisms for *S. herbacea* transport to Baffin Island, Greenland, and Svalbard during the Holocene. North of Iceland, proxy evidence shows that sea ice disappeared by ~11200 BP (Figure 2B [↗](#), Xiao et al., 2017 [↗](#)) when surface ocean currents were established bringing warm Atlantic waters to the North Iceland Shelf (Figure 2C [↗](#), Eiríksson et al., 2000 [↗](#); Sha et al., 2022 [↗](#)), nearly 500 years before the colonization of the first woody taxon on the island (Figure 2A [↗](#)). Thus, with the absence of sea ice around Iceland's coastline during woody plant colonization, wind and/or birds remain the likeliest dispersal mechanisms for Iceland, similar to other Arctic regions.

Future outlook

As the planet continues to warm, glacier and ice sheet mass loss is accelerating (Hugonnet et al., 2021 [↗](#)), providing new territory for woody shrubs to colonize – the Greenland Ice Sheet currently occupies 1.7 million km² (Morlighem et al., 2017 [↗](#)) with >400,000 km² more covered by glaciers elsewhere in the Arctic (RGI Consortium, 2017 [↗](#)). Global warming will also allow woody plants to

migrate poleward across unglaciated Arctic tundra (e.g., Myers-Smith et al., 2011 [↗](#)), which occupies a substantially larger area compared to glaciated regions (5 billion km², Walker et al., 2005 [↗](#)). In this study, we demonstrate that Salicaceae can migrate into deglaciating North Atlantic habitats rapidly whereas Betulaceae may be delayed by millennia. However, the ability of these taxa to successfully follow the rapid pace of ongoing range shifts in non-glaciated tundra (e.g., Alaska and Siberia) remains an open question (e.g., Davis and Shaw, 2001 [↗](#); Thuiller et al., 2009). This is partially due to different baseline conditions of the tundra ecosystem compared to glacial regions upon deglacial warming, which already contain established ecosystems and nutrient-rich soils bound in the permafrost (e.g., Heijmans et al., 2022).

While the causes and rates of present-day climate change differ from the preceding postglacial period (e.g., Tierney et al., 2020 [↗](#)), the factors that we identify as potentially driving the colonization patterns (i.e., environmental tolerance, modes of reproduction, and soil development) are relevant in both scenarios. As a result, circum North Atlantic shrubification in deglaciating regions may be prolonged – ultimately dampening the rate of change, at least temporarily, in regional biodiversity and food web structure, as well as high-latitude temperature amplification. An important next step for constraining future shrubification rates is developing detailed paleoecological reconstructions of plant dynamics in multiple tundra regions (e.g., Clarke et al., 2020 [↗](#); Huang et al., 2021 [↗](#)), continuing these efforts in glaciated regions, and pairing plant histories with independent local records of climate, such as temperature and precipitation. The latter paleoclimate reconstructions will enable more detailed assessment of plant colonization lags relative to climate (e.g., Crump et al., 2019 [↗](#)). By pairing paleoecological and paleoenvironmental records from formerly glaciated and tundra regions and integrating these with predictive models (e.g., Braconnot et al., 2012 [↗](#)), the community will be able to reduce uncertainties and better quantify future shrubification rates, which are essential to inform ecological predictions for policymakers.

Materials and methods

Lake sediment cores and age control

Stóra Viðarvatn (66.24°N, 15.84°W) is a large (2.4 km²), deep lake (48 m) located at 151 m asl in northeast Iceland (**Figure 1** [↗](#), Axford et al., 2007 [↗](#)). In winter 2020, we recovered a composite 8.93 m long sediment core (20SVID-02, Fig. S1) from 17.4 m water depth using lake ice as a coring platform. The sediment was collected in 7 drives of ~150 cm each. The core sections were subsequently stored at 4 °C before opening for sediment subsampling. The chronology of 20SVID-02 is based on 5 marker tephra layers (volcanic ash) of known age: Askja S, G10ka Series, Kverkfjöll/Hekla, Kverkfjöll, and Hekla 4 tephra layers, with ages of 10830 ± 57 BP (Bronk Ramsey et al., 2015), 10400 to 9900 BP (Óladóttir et al., 2020 [↗](#)), 6200 BP (Óladóttir et al., 2011 [↗](#)), 5200 BP (Óladóttir et al., 2011 [↗](#)) and 4200 ± 42 BP (Dugmore et al., 1995 [↗](#)), respectively (see Supplemental Information). A Bayesian age model was generated using the R package rbacon and default settings (Blaauw and Christen, 2011 [↗](#)) (Figure S2A).

Torfdalsvatn (66.06°N, 20.38°W) is a small (0.4 km²), shallow (5.8 m) lake located at 52 m asl in north Iceland (**Figure 1** [↗](#), Rundgren et al., 1997). Paleovegetation records from this lake are based on pollen, macrofossil assemblages (Björck et al., 1992 [↗](#); Rundgren, 1995 [↗](#), 1998 [↗](#); Rundgren and Ingólfsson, 1999 [↗](#)), and *sedaDNA* (Alsos et al., 2021 [↗](#)). Sediment cores from Torfdalsvatn were also recovered by a joint Colorado-Iceland team in 2004, 2008, 2012, and 2020. In modifying the lake sediment core chronologies, we use the recently calibrated age model (Figure S2B, Geirsdóttir et al., 2020 [↗](#)) for the pollen/macrofossil records (Björck et al., 1992 [↗](#); Rundgren, 1995 [↗](#)) that uses the IntCal20 calibration curve (Reimer et al., 2020 [↗](#)). We note that the ~12000 BP Vedde Ash has been previously identified in this record based on major oxide composition (Björck et al., 1992 [↗](#)). However, the age of the Vedde Ash is stratigraphically too old for the

radiocarbon-based chronology, suggesting that this tephra layer likely represents ash from a separate and subsequent volcanic eruption that produced the Vedde Ash, of which several possible correlations have been identified in Iceland and mainland Europe (e.g., [Pilcher et al., 2005](#); [Kristjánsdóttir et al., 2017](#); [Matthews et al., 2011](#); [Geirsdóttir et al., 2022](#)). For the *sedDNA* record, the Early Holocene portion of lake sediment core ([Alsos et al., 2021](#)) features two age control points, the so-called “Saksunarvatn Ash” (10267 ± 89 BP) and a graminoid radiocarbon age (6620 ± 120 BP, [Alsos et al., 2021](#)). To better compare with the pollen/macrofossil record, we generated a new age model for the *sedDNA* record ([Alsos et al., 2021](#)), using the lowermost radiocarbon age and the upper and lower limits of the main ~20 cm thick G10ka Series/Saksunarvatn tephra unit (10400 to 9900 BP) following the recent recognition of multiple associated tephra layers and dating of these layers in Iceland ([Harning et al., 2018](#), [2019](#); [Óladóttir et al., 2020](#)). The new Bayesian age model was generated using the R package *rbacon* and default settings ([Blaauw and Christen, 2011](#)) with the IntCal20 calibration curve ([Reimer et al., 2020](#)) (Figure S2C).

DNA metabarcoding

We subsampled the Stóra Viðarvatn sediment core halves immediately upon splitting in a dedicated clean lab facility with no PCR products in the Trace Metal Lab, University of Colorado Boulder. All surfaces and tools were treated with 10% bleach and 70% ethanol before use. Personnel wearing complete PPE took a total of 75 samples using clean metal spatulas. Samples were then stored in sterile Falcon tubes at 4°C until DNA extraction and metabarcoding. Sampling resolution was based upon a preliminary age model with a sample estimated every 150 years.

We performed DNA extraction and metabarcoding library generation in a dedicated ancient DNA laboratory at the Paleogenomics Lab, University of California Santa Cruz. While working in this laboratory we followed commonly used practices for minimizing contamination of ancient and degraded samples described previously ([Fulton and Shapiro, 2019](#)). We initially cleaned all instruments and surfaces using 10% bleach and 70% ethanol before UV irradiation for at least 1 hour. From each homogenized sample of the core, we aliquoted two 50 mg subsamples and isolated DNA following the column purification method described in [Rohland et al. \(2018\)](#) using Binding Buffer D and eluting 50 µl volumes. We pooled extraction eluates derived from the same original core sample prior to subsequent experiments and included one extraction control (containing no sample) for every 12 samples processed to assess the presence of external contaminants or cross contamination.

We performed quantitative PCR (qPCR) to determine the ideal dilution of extract to add to the metabarcoding PCR, as well as the ideal number of PCR cycles. We added 2 µl of neat or diluted extract to wells containing 1X Qiagen Multiplex master mix, as well as a final concentration of 0.16 µM of the forward and reverse *trnL* primer and 0.12X SYBR Green I Dye. Cycling conditions were as follows: a 15 min 95C activation, followed by 40 cycles of 94C for 30 sec, 57C for 30 sec, and 72C for 60 sec. We tested 1:10 and 1:100 dilutions of extract to water alongside neat extract. We used cycle threshold (C_T) values to compare PCR efficiency between dilutions and determined the ideal cycle number for the metabarcoding PCR as the number of cycles at which the PCR curve reached the plateau phase.

We performed metabarcoding PCR on each extract using the best dilution as determined by qPCR and versions of the chloroplast *trnL* primer set ([Taberlet et al., 2007](#)) that contained attached truseq adapters (Table S2). We used five replicate PCRs for each primer set and for each sample, with reaction conditions as follows: 2 µl of diluted or undiluted extract in 1X Qiagen Multiplex Mix and a 0.16 µM concentration of each primer at a total volume of 25 µl. Thermocycling began with 15 min at 95C, followed by a number of cycles (as determined by qPCR) at 94C for 30 sec, 50-57C for 90 sec, 72C for 60 sec. We used one negative control PCR (no sample) alongside every 16

samples to assess the possibility of contamination during the PCR set up. Extraction controls also underwent metabarcoding. We cleaned reactions using SPRI select beads and eluted them into a final volume of 20 μ l.

We then used indexing PCR to attach the full truseq adapter and unique identifier. We added 5 μ l of cleaned PCR product from the metabarcoding PCR to 1X Kapa Hifi master mix and dual-unique truseq style indices at a concentration of 0.4 μ M. Thermocycling began with an activation step at 98C for 3 min, followed by 8 cycles of 98C for 20 sec, 65C for 30 sec, 72C for 40 sec, and a final extension at 72C for 2 min. We cleaned reactions using SPRI select beads and quantified DNA concentration using the Nanodrop 8000 Spectrophotometer. We pooled PCR replicates from all samples in equimass ratios and sequenced the pools on an Illumina Nextseq 550 platform. Sequencing was performed using Illumina v3 chemistry and a 150 cycle mid output sequencing kit with paired end reads of 76 base pair length.

We processed raw fastq files and performed taxonomic assignment using the Anacapa Toolkit (Curd et al., 2019) and the ArcBorBryo library, which contains 2280 sequences of Arctic and boreal vascular plants and bryophytes (Sønstebo et al., 2010). Briefly, we removed adapter sequences from the raw fastq data using cutadapt and trimmed low quality bases using FastX-toolkit before reads were separated into paired and unpaired files (Hannon, 2010; Martin, 2011). We then used dada2 for further filtering of chimeric sequences and for merging of paired reads (Callahan et al., 2016). We dereplicated identical sequences present in the data using dada2 and generated ASV (amplicon sequence variant) tables and fastq files for each of the read types (paired merged, paired unmerged, unpaired forward, and unpaired reverse). ASV's were assigned taxonomy by globally aligning to a CRUX database generated from the ArcBorBryo library using Bowtie2 before assignment with Anacapa's custom BLCA algorithm (Langmead and Salzberg, 2012). Outputted taxonomy site frequency tables containing taxonomic assignments and read counts were subsequently analyzed using R. ASV sequences for Salicaceae and Betulaceae identified in our samples are included in the supplement (Table S3).

To minimize the chance of false positives, we retained only sequences that: 1) were not detected in negative controls, 2) had 100% match to sequences in the library, 3) had a minimum of 10 reads per PCR replicate and occurred in a minimum of two of five PCR replicates, and 4) had a minimum of 100 reads across all PCR replicates (Table S4). We verified non-native taxa identified by this pipeline via comparison with the NCBI nucleotide database using BLAST (<http://www.ncbi.nlm.nih.gov/blast/>) and removed these due to the low likelihood of correct taxonomic assignment. Finally, DNA quality scores were calculated following Rijal et al. (2021).

Published *sedaDNA* datasets

To place the Icelandic plant *sedaDNA* datasets in the context of the circum North Atlantic, we compiled all existing lake *sedaDNA* records of vascular plants from this region that were formally glaciated during the Last Glacial Maximum (Figure 1, Table 1). These include records from Baffin Island ($n=1$, Crump et al., 2019), northern Greenland ($n=1$, Epp et al., 2015), Svalbard ($n=2$, Alsos et al., 2016; Volstad et al., 2020), northern Norway ($n=3$, Rijal et al., 2021; Alsos et al., 2022), and southern Norway ($n=1$, ter Schure et al., 2021). To ensure that all records were comparable, we excluded 7 records from northern Norway: Sandfjorddalen, Horntjernet, Gaupstjern, Jøkelvatnet, Kuutsjärvi, Nesservatnet, Sierravannet (Rijal et al., 2021; Alsos et al., 2022). Nesservatnet and Sierravannet contained only Late Holocene records and thus do not capture local postglacial colonization patterns. A lack of supporting proxy records (e.g., carbon content) for Sandfjorddalen and Horntjernet meant it was unclear whether these two records capture the complete postglacial period. For Gaupstjern, the inferred basal age (8776 BP) is inconsistent with the reconstructed timing of deglaciation (~11000 BP, Hughes et al., 2016), which suggests the initial postglacial interval may be missing. Finally, both Jøkelvatnet and Kuutsjärvi were impacted by glacial meltwater during the Early Holocene when woody taxa are first identified (Wittmeier et al., 2015; Bogren, 2019), and thus the inferred timing of plant

colonization is probably confounded in this unstable landscape by periodic pulses of terrestrial detritus. Unless otherwise specified, the timing of deglaciation at each site is inferred from the basal age of the sediment record.

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Author contributions

GHM, ÁG and JS conceptualized funded the project; GHM, BS and ÁG supervised the project; GHM, ÁG, DJH and JHR cored the lake; DJH and NA sampled the sediment; TT performed tephra analyses and identification; SS performed DNA extraction and library generation; DJH performed taxonomic assignment and processed the data, constructed age models, and compiled published datasets; DJH wrote the paper with contributions and expertise from SS, KAJ, NA, TT, JS, ÁG, BS and GHM.

Competing interests

The authors declare that they have no competing interests.

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Editors

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Joint Public Review:

The revised version of the manuscript "Delayed postglacial colonization of *Betula* in Iceland and the circum North Atlantic" by Harning et al. investigates the colonization of shrubs during the Late Pleistocene/Holocene in Northern America and Europe by comparing published sedimentary ancient DNA (sedaDNA) records (and pollen data) with a new sedaDNA record from Island. The manuscript aims to identify shrub colonization patterns, discusses their drivers and evaluates the importance of shrubification under future warming.

The revised version improved the clarity of methods and discussion and results presented are more convincing.

However, parts of the methods (e.g. assessment of blanks and data filtering) and results (e.g. visualization of plant community data) could still be polished, and the figures should be improved to increase the clarity of the manuscript.

Author Response

The following is the authors' response to the current reviews.

We thank Reviewer 1 for their time reviewing our revised manuscript and appreciate their thoughtful suggestions for further clarity. In regard to the public review statement, "However, parts of the methods (e.g. assessment of blanks and data filtering) and results (e.g. visualization of plant community data) could still be polished, and the figures should be improved to increase the clarity of the manuscript", we have made small modifications in the text and figures during production of the Version of Record to address these important suggestions.

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

This manuscript compiles the colonization of shrubs during the Late Pleistocene in Northern America and Europe by comparing plant sedimentary ancient DNA (sedaDNA) records from different published lake sediment cores and also adds two new datasets from Island. The major findings of this work aim to illuminate the colonization patterns of woody shrubs (Salicaceae and Betulaceae) in these sediment archives to understand this process in the past and evaluate its importance under future deglaciation and warming of the Arctic.

We greatly appreciate the time and detailed consideration of our manuscript by Reviewer 1. Our responses to individual comments are highlighted in blue, with the original comments provided by the reviewer in black.

The strength of evidence is solid as methods (sedimentary DNA) and data analyses broadly support the claims because the authors use an established metabarcoding approach with PCR replicates (supporting the replicability of PCR and thereby proving the occurrence of Salicaceae and Betulaceae in the samples) and quantitative estimation of plant DNA with qPCR (which defines the number of cycles used for each PCR amplification to prevent overamplification). However, the extraction methods need more explanation and the bioinformatic pipeline is not well-known and needs also some further description in the main text (not only referring to other publications).

Thank you for bringing this to our attention. We have now provided greater detail on our extraction methods and bioinformatic pipeline.

The authors compare their own data with previously published data to indicate the different timing of shrubification in the selected sites and show that Salicaceae occurs always like a pioneer shrub after deglaciation, followed by Betulaceae with a various time lag. The successive colonization of Salicaceae followed by Betulaceae is explained by

its differences in environmental tolerance, the time lag of colonization in the compared records is e.g. explained by varying distance to source areas.

However, there are some weaknesses in the strength of evidence because full sedaDNA plant DNA assessment, quality of the sedaDNA data (relative abundance and richness of sedaDNA plant composition) and results from Blank controls (for sedaDNA) are not fully provided. I think it is important to show how the plant metabarcoding in general worked out, because it is known that e.g. poor richness can be indicative of less preserved DNA and a full plant assessment (shown in the supplement) would be more comprehensive and would likely attract a larger readership.

Thank you for bringing these important points to our attention. The DNA dataset including the full taxa assemblage will be included with the manuscript upon publication and apologize for not including it during the review process. This dataset will also include information on positive and negative blanks used for quality control. Following suggestions from Reviewer 2, we have now also calculated some recently proposed DNA quality metrics (Rijal et al., 2021), which collectively support our earlier conclusions that our record is of sufficient quality to draw the current conclusions. We hope that the inclusion of the complete DNA dataset will indeed draw a larger readership.

Further, it would allow us to see the relative abundance in changes of plants and would make it easier to understand if the families Salicaceae and Betulaceae are a major component of the community signal. Further, the possibility to reach higher taxonomic resolution with sedaDNA compared to pollen or to facilitate a continuous record (which is different from macrofossils) is not discussed in the manuscript but should be added. Also, the taxonomic resolution within these families in the discussed datasets would be of interest, also on the sequence type level if tax. assignments are similar.

Thank you for these suggestions. We have focused on these two families as it is known from numerous pollen records and floras that they are the major component of the vascular plant communities in the regions investigated. *Betula* (birch) and *Salix* (willow) are indeed the most dominant woodland shrubs of the tundra biome, which covers expansive areas of the Arctic. For example, in Iceland natural woodlands, which cover 1.5% of the total land area, are composed of 80% birch shrubs (Snorrason et al. 2016, Náttúrufræðingurinn 86). *Salix* mixes in with *Betula*, especially around wet sites. Species from both genera are common and wide-spread throughout Iceland, but dwarf and cold tolerant species thrive best on the highland or at glacial sites, while shrub-like species are more common on the lowland, coastal area and in sheltered valleys. Flora of Iceland (<http://www.floraislands.is/PDF-skjol/Checklist-vascular.pdf>) lists *Betula* as the only genus of Betulaceae native to Iceland (page 79/80) and *Salix* as the major genus of Salicaceae (page 82-85), although *Populus tremula* (Salicaceae) exists in the wild but is rare (perhaps just a countable number of trees/shrubs in the whole country). The point is that, for Iceland, Betulaceae is *Betula* and Salicaceae is *Salix*, meaning that our sedaDNA method has the taxonomic resolution at the genus level. And with the help of pollen analysis of the site near Stóra Viðarvatn (the novel sedaDNA work of the present paper), i.e., Ytri-Áland site (Karlisdóttir et al. 2014), it is possible to interpret our results even to the species level, which we have only mention in the discussion. It has been suggested that matching sedaDNA results with botanical knowledge about the study site and the vegetation history (local reference database) is one way to increase taxonomic resolution of the sedaDNA approach (e.g. Elliott et al. 2023, Quaternary 6,7). In the same way we find our sedaDNA analysis having sufficient resolution to answer the questions asked in the present study. For the future, although we do not include it in the discussion this time, it should be possible to increase the taxonomic resolution of plant metabarcoding by priming multiple genes simultaneously like that is described as a proof of concept by Foster et al. (2021, Front Ecol Evol 9: 735744). In the revised version of the manuscript, we have now expanded on the

power of sedaDNA in terms of increased taxonomic resolution and application in continuous lake sediment records in the introduction of the manuscript. Following Reviewer 2's suggestion, we have now included the sequences used for taxonomic assignment in the supplement information.

Another important aspect is how the abundance/occurrence of Salicaceae is discussed. Many studies on sedaDNA confirm an overrepresentation of this family due to better preservation in the sediment, far-distance transport along rivers, or preferences of primers during amplification etc. As this family is the major objective of this study, such discussion should be added to the manuscript and data should be presented accordingly.

Thank you for raising this point. The reviewer is indeed correct that Salicaceae is typically overrepresented in read abundance compared to other vascular plant taxa in sedaDNA studies. However, as we mention in the Results and Interpretation section for Stóra Viðarvatn "As PCR amplification results in sequence read abundances that may not reflect original relative abundances in a sample (Nichols et al., 2018), we focus our discussion on taxa presence/absence," we do not place weight on the indeed greater relative abundance of Salicaceae in our own dataset. As such, this different relative abundance of plant taxa reads should not influence the conclusions drawn in the manuscript.

I also miss more clarity about how the authors defined the source areas (refugia) of the shrubs. If these source areas are described in other literature I suggest to show them in a map or so. Further, it should be also discussed and explained more in detail which specific environmental preferences these families have, this is too short in the introduction and too unspecific. Also, it would be beneficial to show relative abundances rather than just highlighted areas in the Figures and it would allow us to see if Salicaceae will be replaced by Betulaceae after colonizing or if both families persist together, which might be important to understand future development of shrubs in these areas.

Thank you for allowing us to clarify. As the regions studied with the lake sediment records shown in this manuscript were all covered by extensive ice sheets during the Last Glacial Maximum (LGM, Fig. 1), plant refugia and source areas must have been located somewhere south of the ice sheet margins. Thus, we calculate our distance to source as the minimum distance from a lake site to land beyond the extent of the ice sheet during the LGM. This has now been clarified in the text and highlighted in Fig. 1. We have also added in the discussion molecular results from Thórsson et al. (2010, J Biogeogr 37) on possible source origins of Betula in Iceland. Details on taxa environmental preferences have now been expanded upon in the Discussion section where we explore the various trait-based factors that may influence the relative differences in colonization timing between Salicaceae and Betulaceae. We have now also edited Figs. 3 and 4 to include PCR replicates instead of highlighted bars to better compare the DNA and pollen datasets from Iceland.

The author started a discussion about shrubification in the future, but a more defined evaluation and discussion of how to use such paleo datasets to predict future shrubification and its consequences for the Arctic would give more significance to the work.

Thank you for this suggestion and allowing us to expand on potential future changes. We have now edited this final section of the paper to provide a little more detail on how we envision these records being used to predict future shrubification and climate change.

Reviewer #1 (Recommendations For The Authors):

I list some more specific details here.

You speak about "read counts", I guess you used relative abundance of read counts, you should state it like this.

Thank you for allowing us to clarify. The data that we refer do in terms of read counts is from the previously published studies in the circum North Atlantic. The data provided from these studies is raw read counts, and not relative abundance.

Line 100: What do you mean here: "temperature changes in prior warm periods"?

Thank you for allowing us to clarify. We have rephrased to sentence to "higher temperature in prior warm periods", which we hope is clearer for the reader.

Line 134: How is DNA diluted by minerogenic sediment? Did the sedimentation rate increase? Typically minerogenic input should be beneficial for DNA preservation.

Thank you for allowing us to clarify. These samples were primarily comprised of tephra glass with minimal organic content. While we agree that minerogenic sediment is generally beneficial for DNA preservation, the predominance of inorganics (tephra) that fell from the sky, rather than being washed into the lake from the landscape, would not carry organic sediment with it. We have rephrased the sentence to make this clearer.

I would suggest adding more citations to the text (for example statements in lines 106, 110, 368)

Thank you for the suggestion. The manuscript has been edited accordingly.

Better divide your discussion part: discussion about dispersal mechanisms occur in both sections. Maybe you could divide it into environmental factors for colonization and traitbased factors (only an idea).

Thank you for the suggestion. We have now edited the second dispersal section to "Environmental dispersal mechanisms" to be clearer about our focus on factors such as wind, sea ice, and birds that may transport the seeds across the North Atlantic. The previous section retains the trait-based factors that may influence relative timing in colonization between Salicaceae and Betulaceae.

Which type of sequencing did you use, paired-end 76bp is unknown to me.

Methods have now been edited to clarify this, along with details related to extraction methods as requested in the Public Review.

Reviewer #2 (Public Review):

Harding et al have analysed 75 sedaDNA samples from Store Vidarvatn in Iceland. They have also revised the age-depth model of earlier pollen, macrofossil, and sedaDNA studies from Torfdalsvatn (Iceland), and they review sedaDNA studies for first detection of Betulaceae and Salicaceae in Iceland and surrounding areas. Their Store Vidarvatn data are potentially very interesting, with 53 taxa detected in 73 of the samples, but only results on two taxa are presented. Their revised age-depth model cast new light on earlier studies from Torfdalsvatn, which allows a more precise comparison to the other

studies. The main result from both sedaDNA and the review is that Salicaceae arrives before Betulaceae in Iceland and the surrounding area. This is a well-known fact from pollen, macrofossil, and sedaDNA studies (Fredskild 1991 Nordic J Bot, Birks & Birks QSR 2014, Alsos et al. 2009, 2016, 2022) and as expected as the northernmost Salix reach the Polar Desert zone (zone A, 1-3°C July temperature) whereas the northernmost Betula rarely goes beyond the Southern Tundra (zone D, 8-9 °C July temperature, Walker et al. 2005 J. Veg. Sci., Elven et al. 2011 <http://panarcticflora.org/>).

We greatly appreciate the time and detailed consideration of our manuscript by Reviewer 2. Our responses to individual comments are highlighted in blue, with the original comments provided by the reviewer in black.

While we agree that previous studies have indeed indicated a relative delay in Betula colonization relative to Salix, most of these have relied on pollen and macrofossil evidence, which are complicated to use as proxies for the first appearance of a given taxa (see our Introduction in the main manuscript). A few studies have shown this also with sedaDNA (e.g., Alsos et al., 2022), which is a more robust proxy for a plant taxa's presence, but these have been limited geographically (e.g., northern Fennoscandia). In our study, we show that this pattern is reflected in 10 different lakes across the North Atlantic, emphasizing the broad nature of Betula's delayed colonization relative to other woody shrubs, such as Salix.

My major concern is their conclusion that lag in shrubification may be expected based on the observations that there is a time gap between deglaciation and the arrival of Salicaceae and between the arrival of Salicaceae and Betulaceae. A "lag" in biological terms is defined as the time from when a site becomes environmentally suitable for a species until the species establish at the site (Alexander et al. 2018 Glob. Change Biol.). The climate requirement for Salicaceae highly depends on species. In the three northernmost zones (A-C), it appears as a dwarf shrub, and it only appears as a shrub in the Southern Tundra (D) and Shrub Tundra (E) zone, and further south it is commonly trees. Thus, Salicaceae cannot be used to distinguish between the shrub tundra and more northern other zones, and therefore cannot be used as an indicator for arctic shrubification. Betulaceae, on the other hand, rarely reach zone C, and are common in zone D and further south. Thus, if we assume that the first Betulaceae to arrive in Iceland is Betula nana, this is a good indicator of the expansion of shrub tundra. Thus, if they could estimate when the climate became suitable for B. nana, they would have a good indicator of colonisation lags, which can provide some valuable information about time lags in shrub expansion (especially to islands). They could use either independent proxy or information from the other species recorded in sedaDNA to reconstruct minimum July temperature (see e.g. Parducci et al. 2012a+b Science, Alsos et al. 2020 QSR).

We appreciate the reviewer's insight into the implications of our use of the word "lag". Indeed, as we do not have site-specific climate timeseries for each lake record, we have adjusted our wording to "delay", which we believe is more general and descriptive of our observations. We recognize the importance of independent paleotemperature records for each lake, but these are not yet available for all records, so we prefer to keep our study focused on the delay instead. In addition, we prefer not to derive temperature records from the vegetation sedaDNA records, as these are not independent and will incorporate changes driven by additional factors, such as soil and light (e.g., Alsos et al., 2022). We have added some text to the final section on Future Outlook that elaborates on the need for complimentary records of past climate to pair with paleoecological records of colonization. We hope that this motivates the community to pursue these lines of research that we agree are needed.

The study gives a nice summary of current knowledge and the new sedaDNA data generated are valuable for anyone interested in the post-glacial colonisation of Iceland. Unfortunately, neither raw nor final data are given. Providing the raw data would allow re-analysing with a more extensive reference library, and providing final data used in their publication will for sure interest many botanists and palaeoecologist, especially as 73 samples provide high time resolution compared to most other sedaDNA studies.

Finally, the raw and final data, including blank controls, used in our study for Stóra Viðarvatn will ultimately be provided with the manuscript's publication. We apologize for not including it with the original submission.

Reviewer #2 (Recommendations For The Authors):

Line 112-113: Difference in northward expansion rate is not the same as lag. Thus, your conclusion "As a result, the biospheres role in future high latitude temperature amplification may be delayed." does not derive directly from the data you present.

Thank you for allowing us to clarify our wording. We have rephrased the sentence to align with our results more closely as stated in the Abstract of the manuscript.

.Line 133: From Figure S3, it looks like three or possibly four samples failed.

Thank you for pointing this out. First, we realized that the DNA reads originally included in Figure S3 were from after filtering. We have now updated the figure to include the total raw reads, which is a better indicator of DNA reliability (Rijal et al., 2021). Based on the total raw reads, only two samples failed with total reads of 2 and 5.

Line 141: You say you focus on presence/absence, but you do show quantitative results for Salix and Betula (0-5 PCR repeats) in Figure 2.

Thank you for allowing us to clarify. Fig 2 shows the number of replicates that meet our criteria for taxa presence, where a higher number of replicates corresponds to a higher likelihood of presence.

Line 142: Where are the other 51 taxa shown?

We are providing the full DNA record in the supplement, which will be published alongside the main manuscript. We have also now included a plot of species richness against sample depth in Fig. S2.

Line 178-179: Note that the revised date of first detection is close to what has been previously published (Salix ~10300 vs. 10227, Betula ~9500 vs 9680), so it does not make any changes to previous interpretation.

Yes, this is true. However, we still believe it is important to always consider improvements in age models to best correlate the timing of events between different paleo records.

Line 191-194 and Figure S2: I leave the evaluation of revised age-depth model to the geologist.

As this aspect was not commented on, we assume that both reviewers are satisfied.

Line 197: "Delay" is a more correct word than "lag".

Thank you, edited.

Line 210: Where do 1700 and 2500 come from? If your revised age of ice retreat is 11 800, and your revised date of Salix and Betula arrival are ~10 300 and ~9500, I make this 1500 and 2300.

Yes, this is correct. Thank you for pointing out this error.

Line 215-217: To be more certain about any bias caused by low DNA quality, I suggest you explore your data using the tools presented in Rijal et al. 2021 Science Advances. As you do not provide your data, I cannot evaluate the quality of them.

Thank you for the suggestion. We have now calculated the various DNA quality indices developed by Rijal et al. (2021). This has been added to the methods and results section for the Stóra Viðarvatn record, as well as in Fig. S3. The MTQ and MAQ scores are known to correlate with species richness when richness is low ($n < 30$, Rijal et al., 2021), which is likely an artifact of the requirement that the 10 best represented barcode sequences are required to calculate these scores. As this correlation is observed in our dataset and given that our species richness is low ($n < 30$, Fig. S2), the low MTQ and MAQ score are not likely indicative of low-quality DNA. We therefore judge the quality of our DNA on total raw reads and CT values, which remain relatively constant through time (Fig. S2).

Line 226: Do you mean TDV?

We intended to omit unnecessary abbreviations throughout the manuscript, such as lake names, in our original manuscript. We have now changed TORF, which we use as the lake's abbreviation, to the full lake name, Torfdalsvatn.

Line 282-283: Given that the basal sediments of Nordivatnet are marine (Brown et al. 2022 PNAS Nexus), even a low detection may be a strong indication of local presence.

Thank you for this point. However, to standardize the records and compare across a wide range of geographical and depositional settings, we prefer to apply the same criteria for the taxa's presence to each lake as outlined in our Methods.

Line 289: See the definition of "lag"

Changed to "delayed" per your earlier suggestion. Thank you.

Line 298-303: I agree that the late appearance of Betula at Langfjordvatnet (10 000 cal BP) is anomalously long and a bit unexpected given that it is found at five other lakes in the region 13000-10200 cal BP (Alsos et al. 2022). However, a likely explanation is the lack of area with stable soil - B. nana requires a greater degree of soil development compared to other heath shrubs (Whittaker 1993) and Langfjordvatnet is surrounded by steep scree slopes (Otterå 2012 master thesis Univ. Bergen). At Jøkelvatnet, Salix appears in the four available samples from 10453 to 9811 whereas Betula arrives 9663. Here, the arrival of Betula is just at the drop of local glacier activity and at the temperature rise, suggesting that it arrives immediately after the climate becomes suitable (Elliott et al. 2023 Quaternary). Thus, based on N Fennoscandia where we have more data available, it does not show lags and does not support delayed shrubification (which contrasts with what we have shown for many other species including common dwarf shrubs, see Alsos et al. 2022). Would be very interesting to have similar data from Iceland, which has a large dispersal barrier.

Thank you for these further considerations. We have incorporated those related to Langfjordvannet into the manuscript accordingly. We also appreciate the point regarding Jøkelvatnet. However, as stated in our Methods section for “Published sedaDNA datasets”, we do not include Jøkelvatnet in our comparison due to the impact of glacier activity as the reviewer notes: “Finally, both Jøkelvatnet and Kuutsjärvi were impacted by glacial meltwater during the Early Holocene when woody taxa are first identified (Wittmeier et al., 2015; Bogren, 2019), and thus the inferred timing of plant colonization is probably confounded in this unstable landscape by periodic pulses of terrestrial detritus.” Due to the glacier’s presence in the lake catchment, it is not possible to discern whether delay in Betulaceae would have occurred if the glacier were not present. Therefore, we prefer to keep this record excluded from our comparisons.

Line 316-319 and 344: Based on contemporary genetic patterns, Alsos et al. analyse the relative importance of adaptation to dispersal compared to other factors.

Thank for you bringing up this important point. We have now expanded our discussion to include these analyses from Alsos et al. (2022).

Line 342+350: Original publication is Alsos et al. 2007 Science

Thank you, edited.

Line 343: Alsos et al. 2009 Salix study is the wrong citation here. Eidesen et al. 2015 Mol. Ecol. shows phylogeography of Greenland population but not Baffin - I am not aware of any contemporary genetic studies of Betula from Baffin.

Thank you for pointing this out. We will also include the Eidesen et al. (2015) citation for reference to Greenland. However, there is one data point included for southern Baffin Island in Alsos et al. (2009), so we will retain this citation here as well.

Line 351-353: See comment about Betula from Baffin above. Also, I am not sure I follow here - what do you mean by "these populations" - the Svalbard ones or Iceland? Eidesen et al. 2015 is the wrong citation for Salix - use Alsos et al. 2009. Alsos et al. 2009 suggest Iceland (and E Grenland) was colonized from north Scandinavia, although this was uncertain as no data were available from Faroe/Shetland. Svalbard was colonized from N Fennoscandia (Alsos et al. 2007).

Regarding Baffin Island sources, we refer the reviewer to our response to their previous comment. We have clarified the wording of our sentence from “these populations” to “the modern populations from these locations [Baffin Island, Greenland, and Svalbard]”. We have removed reference to Eidesen et al. (2015), as this is for Betula rather than Salix. Finally, we have added a citation for Alsos et al. (2007) here for Svalbard.

Line 354-355: AFLP suggest that Baffin and W Greenland were colonised from a refugia south of the Wisconsin Ice Sheet, see Alsos et al. 2009.

Yes, we are aware, thank you. Our reference to “mid-latitude North America” in the sentence acknowledges this refugia, but we have now added “south of the Laurentide Ice Sheet” for further clarification.

Line 363-381: See comment above; your Store Vidarvatn data do currently not demonstrate a lag between environmental suitability and climate, but using the rest of

the DNA record, potentially it could. Would also be good to reflect on the distance to the source area for shrubs Late Glacial/Early Holocene compared to now.

Thank you for these suggestions. We have edited this section of the manuscript to elaborate on the need for independent climate reconstructions as well as the fact that distances to plant refugia are shorter now than during the last postglacial period.

Line 396-416: I am not an expert on tephra so I will not comment on this part.

As this aspect was not commented on, we assume that both reviewers are satisfied.

Line 459-457: Please provide results of how much data is lost at each step of filtering.

We added the read loss following each filtering step as a table in the supplemental information (Table S4).

Throughout the manuscript, you go only to species level although DNA in most cases is able to distinguish to genus level within Salicaceae and Betulaceae - which sequences did you identify?

Sequences are now provided in the supplemental for Salicaceae and Betulaceae. Based on our bioinformatic pipeline, reference library and requirement for 100% match between sequence and taxonomy, we were only able to distinguish between species level.

Figure 2: The detection of Betulaceae is very sporadic in Stóra Vidarvatn with occurrence in only seven samples and hardly ever in all 5 repeats, suggesting that if you apply a statistical model to estimate first arrival (see Alsos et al. 2022), you will have a large confidence interval. Thus, these uncertainties should be considered when estimating the delayed arrival of Betula compared to Salix. The data from Torfdalsvatn (which I assume are from Alsos et al. 2021 although not specified in the figure legend), shows detection in all samples from the first appearance and mostly in 8 of 8 repeats (shown in the original publication - you could do the same here), thus providing a more accurate estimate for the time gap between arrival of Salix and Betula.

Thank you for bringing up this important point. The detection of Betulaceae is indeed sporadic, but we believe it reflects the genuine nature of its presence/absence during the Holocene in Northeast Iceland. This is supported by Betula pollen from a nearby peat record that shows a similar history (Fig. 4, Karlsdóttir et al., 2014), which we have now elaborated on in the Results and Interpretation section. As for the timing of Betulaceae colonization at this site, the first appearance in the DNA record should be a close minimum estimate as shown with modern DNA and plant survey comparisons (e.g., Sjögren et al., 2017; Alsos et al., 2018). The true first appearance could be biased by small amounts of plants being present in the early stages of colonization and not registering the sedimentary record until enough dead plant material is transported to the depocenter of the lake. However, this is likely less than age model uncertainties and therefore not likely relevant on geologic timescales as in this study. In this sense, our age models and those published for the other records indicate this is generally on the order of several hundred years. In addition, we have now added the Alsos et al. (2021) reference for Torfdalsvatn. Unfortunately, this Torfdalsvatn study does not provide number of PCR repeats so we will keep the figure as is as it best represents the available data.

Figure 5: I suggest adding lake names to the figure. Is there a dot missing for lake 5 for Salicaceae?

Thank you for the suggestion, we have added lake names to the figure. There is a dot marked for Salicaceae for lake 5, however, not for Betulaceae as this taxon was not identified. We refer the reviewer to the Discussion Section “Postglacial sedaDNA records from the circum North Atlantic” and the lake’s original publication (Volstad et al., 2020).

Figure 6: I find it more relevant to plot colonization time versus distance to LGM sheetice margin - lake number is just an arbitrary number.

We appreciate the suggestion and have modified the figure accordingly.